

# Supplementary material from "An Entropy-based Framework for Hybrid Coalitions in Game Theory. Part I: Human arbitration"

**Chapter II:** from  $H = 15,000$  to  $H = 50,000$

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## 1 Introduction

This second chapter continues the finite-horizon analysis presented in Chapter I and considers the independently executed long horizons  $H \in \{15000, 30000, 50000\}$ . All theoretical definitions, simulation conventions, state-generation mechanisms, learning-rate schedules, and graphical quantities remain those described in the main paper and in Chapter I and are not repeated here [1].

Whereas Chapter I documented the exploratory, early-learning, and emerging-stabilization stages, the present chapter examines whether the observed Human–AI alignment persists as the available interaction history increases. Particular attention is given to the distinction between genuine two-state alignment and residual schedule- or coverage-dependent discrepancies that may remain visible at long horizons.

The results retain the same organization by policy trajectories, utility-credit traces and finite-time value iterates, and Jensen–Shannon divergence with cumulative execution frequencies. Figures are selected to emphasize the most informative long-horizon comparisons, while the complete numerical outputs and nondisplayed trajectories remain available in the supplementary repository.

The graphical analysis follows the same methodology, notation, and interpretative conventions used in Chapter I. Each horizon is examined through three complementary dimensions:

1. **Policy trajectories.** These plots show the evolution of the predecision Human and AI policies in states  $A$  and  $B$ . The displayed curves report the probabilities assigned to action  $a_0$ :  $p_H(a_0 | A)$ ,  $p_{AI}(a_0 | A)$ ,  $p_H(a_0 | B)$ , and  $p_{AI}(a_0 | B)$ . The corresponding  $a_1$  probabilities are not repeated because they are their exact binary complements.
2. **Utility-credit traces and value iterates.** These plots report the EWMA utility-credit traces  $\hat{U}_H$  and  $\hat{U}_{AI}$ , the derived coalition trace  $\hat{U}_{\text{coal}}$ , and the finite-time computational value iterates. As in Chapter I, these quantities are interpreted as diagnostics of the realized simulation path, rather than as structural utilities or theoretical fixed-point values.
3. **Jensen–Shannon divergence and cumulative execution frequencies.** These plots show the evolution of  $D_{JS}^T$  relative to the delegation thresholds, together with the cumulative empirical Human and AI execution frequencies. They describe jointly the contraction of policy discrepancy and the stabilization of realized execution authority.

To avoid repetition, the definitions of these quantities are not restated in this chapter. The analysis instead concentrates on their long-horizon behaviour, the persistence of two-state alignment, and any remaining dependence on the learning schedule or state coverage.

## 2 Advanced alignment: $H = 15,000$ iterations

At  $H = 15000$ , the selected trajectories exhibit mature Human–AI alignment and strongly AI-dominant execution. The horizon is nevertheless treated as an advanced-alignment regime rather than as universal stationarity: the complete grid still reveals sensitivity of the global diminishing schedule to strongly imbalanced exogenous state visitation.

### 2.1 Policy trajectories

#### 2.1.1 Constant- $\eta$ learning

Under endogenous state generation, the Human and AI policies become practically aligned in both states by approximately  $T = 298$  and numerically aligned by  $T = 467$ . Alignment therefore precedes the late movement of the common policies toward the boundary: both state-specific probabilities continue to evolve after the inter-agent discrepancy has become negligible and reach  $p(a_0 | A) = p(a_0 | B) = 1$  only near the end of the horizon. The selected trajectory spends 98.73% of the decisions in the agreement region, and the AI executes 99.91% of the actions.

The exogenous configuration with  $p_A = 0.90$  provides an even clearer separation between alignment and policy stationarity. State  $A$  approaches the pure action- $a_0$  boundary, while the aligned common policy in the less frequently visited state  $B$  remains mobile and ends at  $p(a_0 | B) = 0.45855$ . Nevertheless, the terminal Human and AI policies coincide in both states, the agreement frequency is 99.19%, and AI execution reaches 99.82%. Thus,  $D_{JS} = 0$  identifies inter-agent alignment, not immobility of the shared state-conditioned policy.

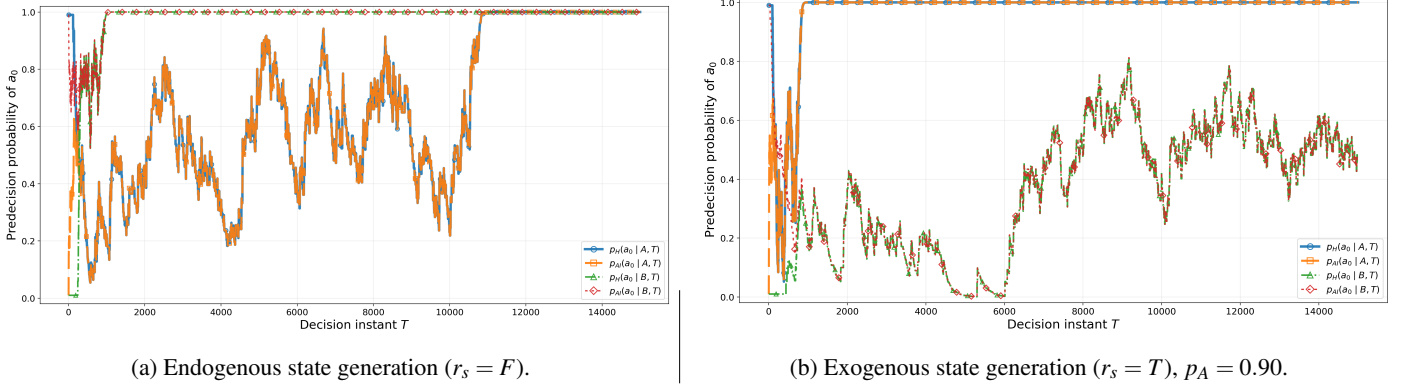


Figure 1: Predecision Human and AI probabilities assigned to  $a_0$  under constant- $\eta$  learning at  $H = 15000$ , with  $\eta = 0.05$ . The endogenous configuration uses  $(\alpha_{\text{agree}}, \alpha_{\text{disagree}}, \beta, \gamma) = (0.35, 0.85, 0.05, 0.50)$ ; the exogenous configuration uses  $(0.20, 0.51, 0.10, 0.10)$ .

### 2.1.2 Global diminishing- $\eta$ learning

The endogenous global-diminishing trajectory reaches practical two-state alignment by approximately  $T = 77$  and numerical alignment by  $T = 260$ . The Human and AI curves then remain superposed within each state, while the two common state-conditioned policies continue a slow and differentiated evolution. They end at  $p(a_0 | A) = 0.13957$  and  $p(a_0 | B) = 0.37673$ . This is a direct long-horizon illustration that Robbins–Monro attenuation can gradually reduce policy movement without making alignment and freeze-out synonymous. The selected run has  $f_{\text{agree}} = 0.99947$  and  $f_{\text{AI}} = 0.99947$ .

With exogenous state generation and  $p_A = 0.30$ , both policies align rapidly in both states and subsequently approach the pure  $a_0$  boundary. Practical and numerical alignment are established by approximately  $T = 176$  and  $T = 331$ , respectively. The flatter long-run segment is consistent with the stronger cumulative reduction of the global step size, while the balanced state coverage avoids the rare-state difficulty observed at extreme values of  $p_A$ .

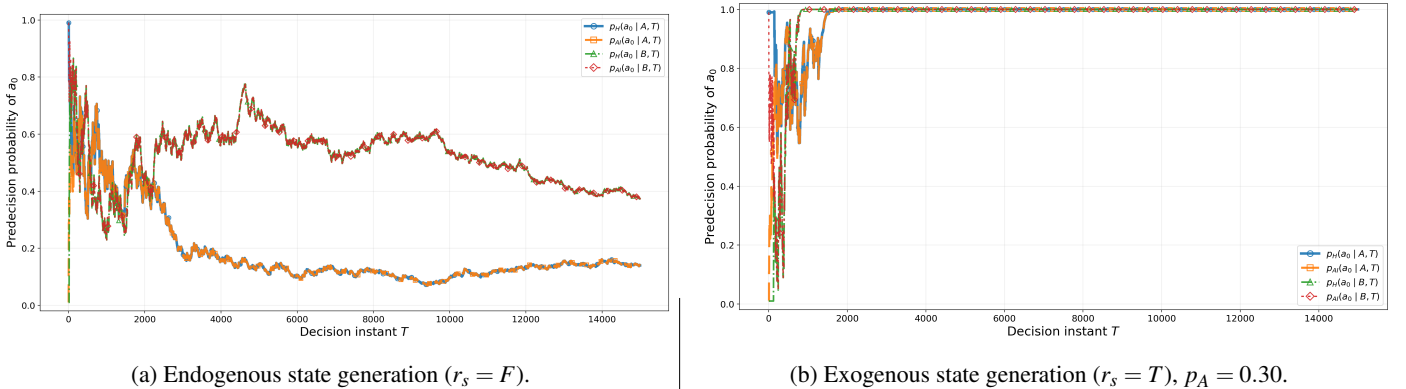


Figure 2: Predecision Human and AI probabilities assigned to  $a_0$  under global diminishing- $\eta$  learning at  $H = 15000$ . The endogenous configuration uses  $\eta_T = 0.08/(1 + 0.002T)$  and  $(\alpha_{\text{agree}}, \alpha_{\text{disagree}}, \beta, \gamma) = (0.49, 0.51, 0.10, 0.50)$ ; the exogenous configuration uses  $\eta_T = 0.10/(1 + 0.001T)$  and  $(0.20, 0.85, 0.02, 0.25)$ .

The exact-empirical policy panels are not repeated at this scale. In the selected endogenous run, practical and numerical alignment occur by approximately  $T = 4$  and  $T = 6$ ; in the exogenous run with  $p_A = 0.99$ , the rare state first appears at  $T = 60$ , followed by practical alignment at  $T = 61$  and numerical alignment at  $T = 183$ . Both selected trajectories remain exactly aligned thereafter. Their near-flat long-horizon plots add little beyond the shorter-horizon visualizations, but the complete figures remain in the repository.

## 2.2 Utility-credit traces and value iterates

The selected utility-credit panels contrast the two most informative long-horizon policy configurations. In the constant- $\eta$  exogenous run, state  $A$  accounts for 90% of the realized path and the AI executes 99.82% of the decisions. The final realized credit traces are  $\hat{U}_H \simeq 0$  and  $\hat{U}_{AI} = \hat{U}_{\text{coal}} = 0.98057$ . With  $\gamma = 0.10$ , the finite-time value iterates remain comparatively small, ending at  $\hat{V}(A) = 1.22 \times 10^{-4}$  and  $\hat{V}(B) = 0.01111$ . These quantities diagnose the realized credit and Bellman paths; they do not replace the structural utilities or the policy-alignment criterion.

The endogenous global-diminishing run has a different state and reward path. AI execution reaches 99.95%, while the final realized AI and coalition credit levels are 0.32297. The corresponding value iterates end at  $\hat{V}(A) = 1.48301$  and  $\hat{V}(B) = 1.73537$ . Their larger scale is consistent with the different discount factor,  $\gamma = 0.50$ , and should not be interpreted as evidence that this configuration is more aligned than the constant- $\eta$  case.

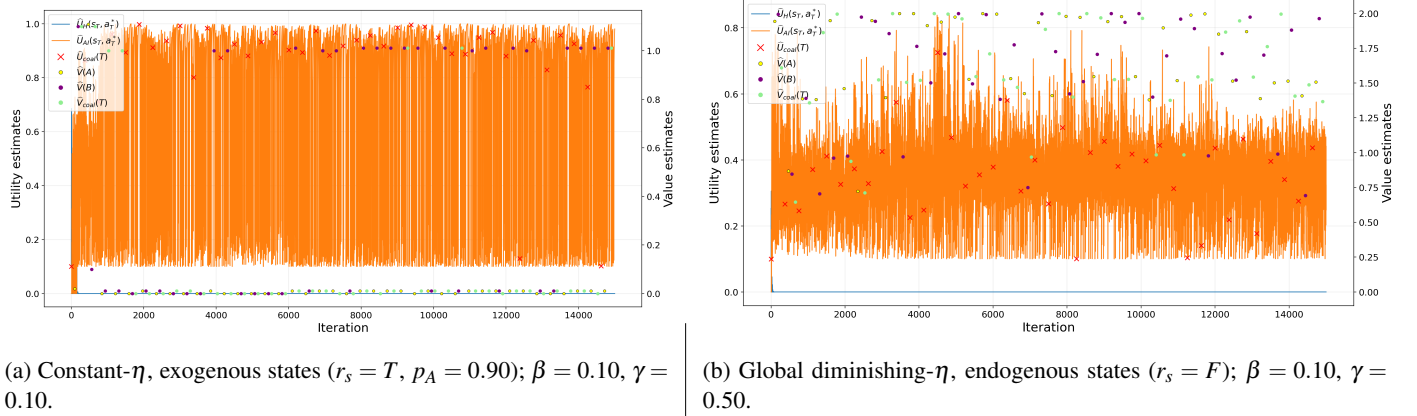


Figure 3: Selected utility-credit traces and finite-time value iterates at  $H = 15000$ . The panels are retained because their policy trajectories remain informative after alignment; the visually redundant exact-empirical and rapidly flattened global-diminishing exogenous panels are reported in the complete repository.

## 2.3 Jensen–Shannon divergence and execution frequencies

### 2.3.1 Constant- $\eta$ learning

In both constant- $\eta$  configurations, the state-conditioned divergence contracts to numerical zero well before the end of the horizon. The endogenous run retains a longer visible contextual transient, with

$$(f_{\text{agree}}, f_{\text{ctx}}, f_{\text{disagree}}) = (0.98733, 0.01253, 0.00013),$$

whereas the exogenous run yields

$$(0.99187, 0.00733, 0.00080).$$

The final execution shares are respectively  $f_{AI} = 0.99907$  and  $f_{AI} = 0.99820$ . The small differences in occupation frequencies therefore arise from the early transient, not from a persistent long-run separation of the Human and AI policies.

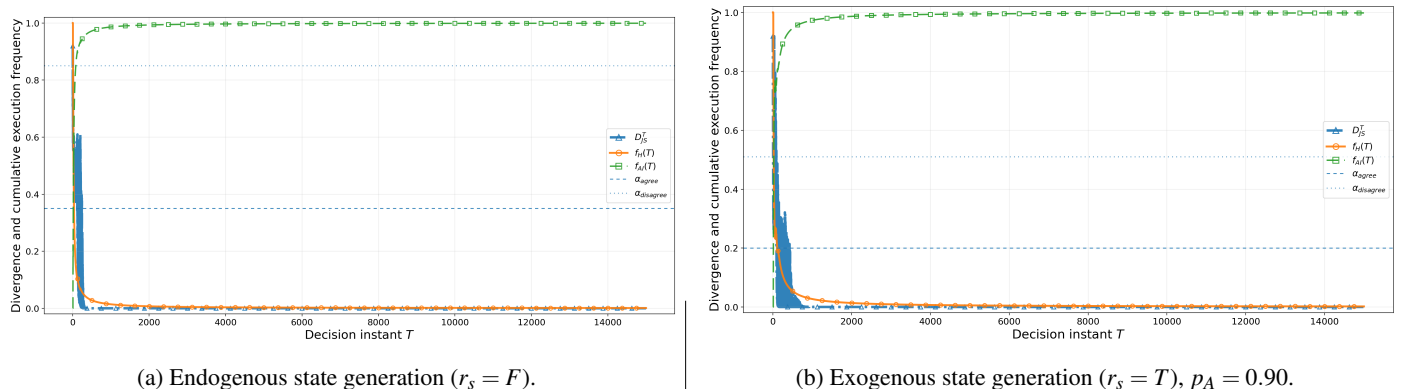


Figure 4: Jensen–Shannon divergence, delegation thresholds, and cumulative execution frequencies under constant- $\eta$  learning at  $H = 15000$ .

### 2.3.2 Global diminishing- $\eta$ learning

The selected global-diminishing runs also end at exact two-state alignment. Under endogenous state generation, the transition is particularly short:  $f_{\text{agree}} = f_{AI} = 0.99947$ , with the residual 0.00053 assigned to disagreement and Human execution. In the exogenous configuration, agreement occupies 99.20% of the path, the contextual region 0.79%, and disagreement 0.01%; AI execution reaches 99.90%. The near-horizontal cumulative frequency curves confirm stabilization of realized execution authority, while the corresponding policy plots show that the aligned common policy may still evolve.

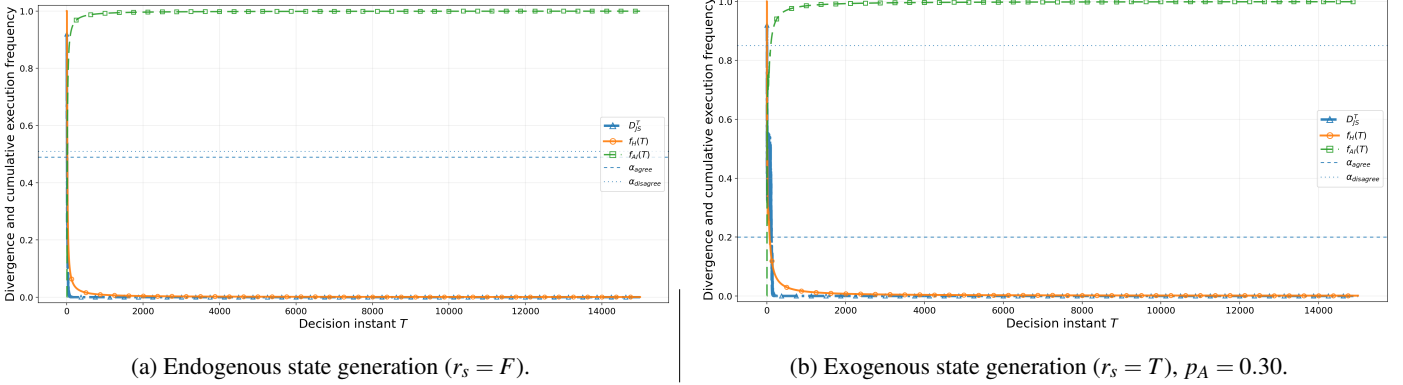


Figure 5: Jensen–Shannon divergence, delegation thresholds, and cumulative execution frequencies under global diminishing- $\eta$  learning at  $H = 15000$ .

## 2.4 Horizon-level synthesis

Table 1: Two-state alignment over the complete  $H = 15000$  grid. Practical alignment requires  $D_{JS,\max}^{\text{final}} \leq 10^{-2}$ , and numerical alignment requires  $D_{JS,\max}^{\text{final}} \leq 10^{-6}$ .

| State generation | Learning schedule          | Practical | Numerical |
|------------------|----------------------------|-----------|-----------|
| Endogenous       | Constant- $\eta$           | 100.00%   | 100.00%   |
|                  | Global diminishing- $\eta$ | 96.67%    | 96.67%    |
|                  | Exact empirical            | 100.00%   | 100.00%   |
| Exogenous        | Constant- $\eta$           | 95.42%    | 87.50%    |
|                  | Global diminishing- $\eta$ | 67.50%    | 57.36%    |
|                  | Exact empirical            | 100.00%   | 100.00%   |

The nondisplayed grid configurations confirm that the selected trajectories are representative of advanced but non-universal alignment. Constant- $\eta$  learning is practically aligned throughout the endogenous grid and in 95.42% of the exogenous grid. Exact empirical updating achieves practical and numerical alignment in every evaluated configuration. Global diminishing learning remains the principal residual exception under exogenous state generation: practical alignment reaches 95% for  $p_A = 0.30$  and  $p_A = 0.70$ , but falls to 25.83% and 24.17% for  $p_A = 0.01$  and  $p_A = 0.99$ , respectively.

The difference follows from the interaction between global time decay and state coverage. A rare state may receive its informative updates only after the global step size has already become small. Constant gain retains nonvanishing responsiveness to such late visits, whereas exact empirical updating uses the local agent–state count and therefore avoids the same global-time attenuation. Consequently,  $H = 15000$  documents consolidated Human–AI alignment and stable AI-dominant execution, while still preserving a measurable dependence on the learning schedule and the visitation frequency of each state.

## 3 Long-horizon consolidation: $H = 30,000$ iterations

At  $H = 30000$ , the principal result is consolidation rather than a new universal regime. Constant- $\eta$  and exact empirical updating retain the strong alignment already observed at  $H = 15000$ , while the global diminishing schedule remains the most informative long-horizon case because it separates Human–AI alignment from the slower movement of the common policy. The displayed panels therefore focus on this schedule; the remaining configurations are reported numerically and remain available in the complete repository.

### 3.1 Policy trajectories

Under endogenous state generation, practical two-state alignment is established by approximately  $T = 106$  and numerical alignment by  $T = 257$ . The Human and AI curves then remain superposed, although the common state-conditioned policies continue to move for several thousand decisions. They enter and remain within 0.01 of their terminal levels only near  $T = 7048$ , ending at  $p(a_0 | A) = 1$  and  $p(a_0 | B) = 0.00370$ . Thus, the reduction of inter-agent divergence occurs well before the attenuation of the common policy trajectory.

The exogenous configuration with  $p_A = 0.30$  makes this distinction still clearer. Practical and numerical alignment occur by approximately  $T = 181$  and  $T = 470$ , respectively, yet the two aligned policies continue a slow joint displacement over almost the entire horizon. They end at the interior state-conditioned probabilities  $p(a_0 | A) = 0.95212$  and  $p(a_0 | B) = 0.85432$ , and remain within 0.01 of these terminal values only near  $T = 29651$ . Hence,  $D_{JS} = 0$  indicates equality of the Human and AI policies, not pointwise stationarity of the policy shared by both agents.

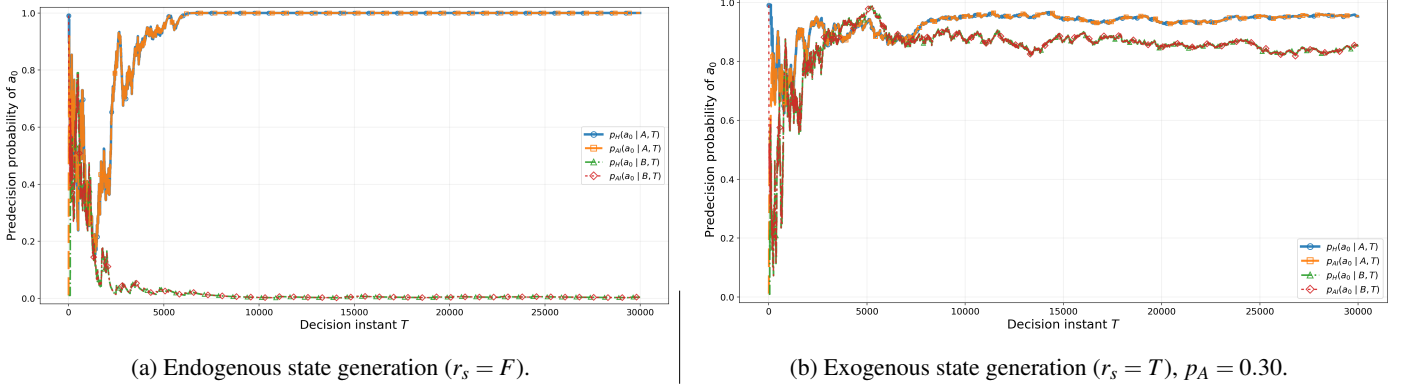


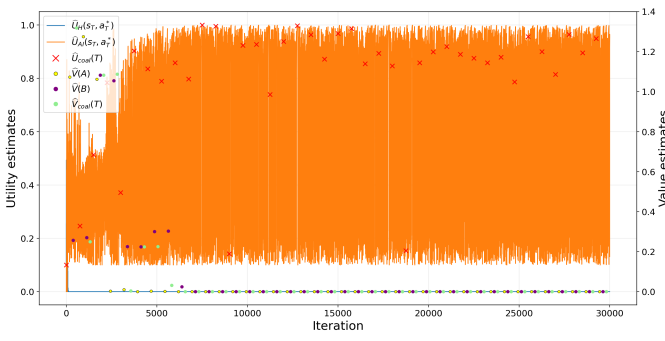
Figure 6: Predecision Human and AI probabilities assigned to  $a_0$  under global diminishing- $\eta$  learning at  $H = 30000$ . The endogenous configuration uses  $\eta_T = 0.10/(1 + 0.002T)$  and  $(\alpha_{\text{agree}}, \alpha_{\text{disagree}}, \beta, \gamma) = (0.20, 0.35, 0.10, 0.25)$ ; the exogenous configuration uses  $\eta_T = 0.08/(1 + 0.002T)$  and  $(0.20, 0.35, 0.10, 0.75)$ .

The nondisplayed constant- $\eta$  winners also reach exact final alignment in both states. Their transitions are concentrated near the origin and the selected policies become effectively fixed by approximately  $T = 1298$  under endogenous generation and  $T = 342$  under exogenous generation with  $p_A = 0.70$ . The exact empirical winners align by approximately  $T = 6$  and  $T = 11$ , respectively, so their informative transients are no longer visible at this horizontal scale. Their omission is therefore graphical rather than substantive.

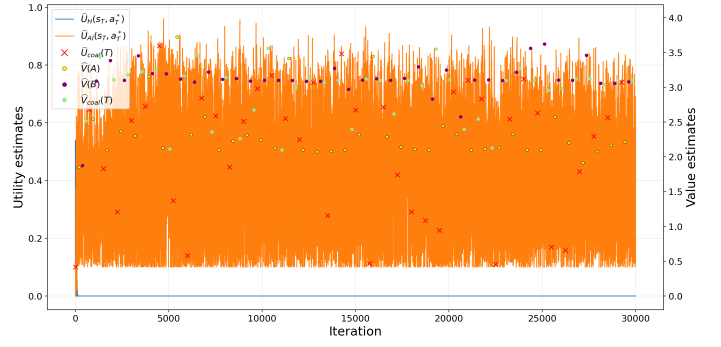
### 3.2 Utility-credit traces and value iterates

The utility-credit panels remain strongly path dependent and visually dense at this horizon. Under endogenous state generation, the AI executes 99.93% of the decisions and the final realized AI and coalition credit traces equal 0.13148. The corresponding finite-time values are numerically negligible at the terminal instant. The repeated switching among realized state-action entries explains the broad vertical variation of the AI credit trace; it does not imply renewed policy divergence.

In the exogenous configuration, the AI executes 99.91% of the decisions and the final AI and coalition credit traces equal 0.60340. The value iterates end at  $\hat{V}(A) = 3.10485$  and  $\hat{V}(B) = 3.32961$ . Their larger magnitude reflects the different discount factor  $\gamma = 0.75$ , rather than a higher degree of policy alignment. These panels are retained as finite-time computational diagnostics, while the shorter horizons remain more informative for visualizing their initial adjustment.



(a) Endogenous state generation ( $r_s = F$ );  $\beta = 0.10$ ,  $\gamma = 0.25$ .



(b) Exogenous state generation ( $r_s = T$ ),  $p_A = 0.30$ ;  $\beta = 0.10$ ,  $\gamma = 0.75$ .

Figure 7: Utility-credit traces and finite-time value iterates for the selected global diminishing- $\eta$  configurations at  $H = 30000$ .

The constant- $\eta$  and exact empirical utility-credit panels are not reproduced because their policy transitions are already compressed at the beginning of the horizon and their diagnostic traces add no new long-horizon mechanism. Their final values and complete trajectories remain available in the numerical files and repository.

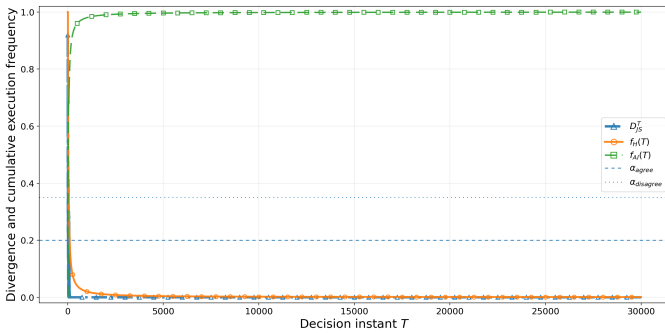
### 3.3 Jensen–Shannon divergence and execution frequencies

The endogenous global-diminishing run spends 99.873% of the horizon in the agreement region, 0.087% in the contextual region, and 0.040% in disagreement. The AI executes 99.933% of the actions. The cumulative curves therefore stabilize rapidly, even though the aligned common policy continues its slower movement for several thousand decisions.

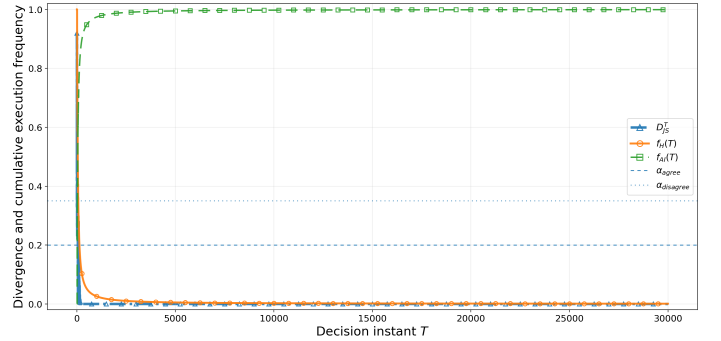
For the exogenous configuration,

$$(f_{\text{agree}}, f_{\text{ctx}}, f_{\text{disagree}}) = (0.99800, 0.00153, 0.00047),$$

and  $f_{AI} = 0.99913$ . The agreement and execution frequencies are therefore effectively stabilized long before the common policies enter their terminal 0.01-neighbourhood. This trajectory provides the clearest  $H = 30000$  evidence that frequency stabilization, policy alignment, and common-policy stationarity are related but distinct finite-horizon properties.



(a) Endogenous state generation ( $r_s = F$ ).



(b) Exogenous state generation ( $r_s = T$ ),  $p_A = 0.30$ .

Figure 8: Jensen–Shannon divergence, delegation thresholds, and cumulative execution frequencies under global diminishing- $\eta$  learning at  $H = 30000$ .

The nondisplayed constant- $\eta$  and exact empirical winners also end with zero state-conditioned divergence. Their agreement frequencies range from 0.99393 to 0.99993, and their AI execution frequencies range from 0.99910 to 0.99993. Consequently, all six selected configurations are AI-dominant in occupation frequency, although the complete grid still distinguishes the learning schedules.

### 3.4 Horizon-level synthesis

Table 2: Two-state alignment over the complete  $H = 30000$  grid. Practical alignment requires  $D_{JS,\max}^{\text{final}} \leq 10^{-2}$ , and numerical alignment requires  $D_{JS,\max}^{\text{final}} \leq 10^{-6}$ .

| State generation | Learning schedule          | Practical | Numerical |
|------------------|----------------------------|-----------|-----------|
| Endogenous       | Constant- $\eta$           | 100.00%   | 100.00%   |
|                  | Global diminishing- $\eta$ | 96.67%    | 96.67%    |
|                  | Exact empirical            | 100.00%   | 100.00%   |
| Exogenous        | Constant- $\eta$           | 97.08%    | 95.42%    |
|                  | Global diminishing- $\eta$ | 69.58%    | 60.28%    |
|                  | Exact empirical            | 100.00%   | 100.00%   |

Relative to  $H = 15000$ , the complete grid changes only modestly. Constant- $\eta$  exogenous practical alignment rises from 95.42% to 97.08%, and global diminishing exogenous alignment rises from 67.50% to 69.58%. The latter improvement remains small because extreme state imbalance continues to expose the interaction between global time decay and rare-state learning. Practical alignment under global diminishing updates reaches 96.67% for  $p_A = 0.30$  and 95.00% for  $p_A = 0.70$ , but only 33.33% for  $p_A = 0.01$  and 26.67% for  $p_A = 0.99$ .

By contrast, exact empirical updating remains practically and numerically aligned throughout the complete endogenous and exogenous grids. Constant gain also benefits from the longer horizon because late observations retain a nonvanishing update magnitude. Therefore,  $H = 30000$  confirms long-horizon consolidation, but not schedule-independent saturation: the principal residual limitation remains the combination of global diminishing gain and severely imbalanced exogenous state visitation.

## 4 Consolidation with residual schedule dependence: $H = 50,000$ iterations

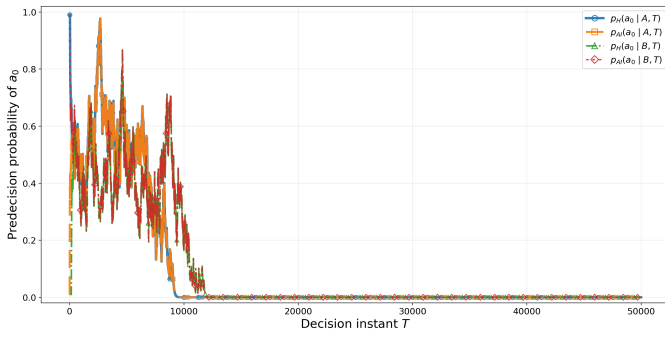
The longest independently executed horizon confirms the persistence of Human–AI alignment and AI-dominant execution, but it does not support a claim of universal saturation. Constant- $\eta$  and exact empirical updating are almost uniformly aligned over the complete grid, whereas the global diminishing schedule retains measurable sensitivity to extreme exogenous state imbalance. The displayed trajectories therefore contrast a rich constant- $\eta$  endogenous path with the most informative global diminishing exogenous case.

### 4.1 Policy trajectories

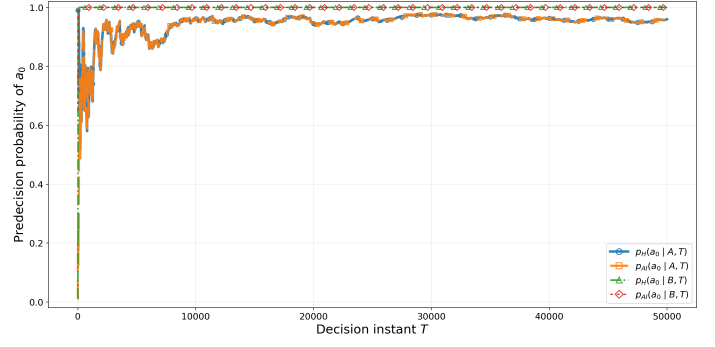
Under constant- $\eta$  learning with endogenous state generation, practical two-state alignment is established by approximately  $T = 274$  and numerical alignment by  $T = 641$ . The aligned policies nevertheless continue to move for a substantial part of the horizon and enter their terminal 0.01-neighbourhood only near  $T = 11885$ . Both state-conditioned probabilities ultimately approach the action- $a_1$  boundary,  $p(a_0 | A) \simeq p(a_0 | B) \simeq 0$ . The trajectory therefore shows that a nonvanishing gain can preserve long-run adaptability after inter-agent discrepancy has already disappeared.

The global diminishing exogenous configuration with  $p_A = 0.30$  provides the strongest separation between alignment and common-policy stationarity in the complete study. Practical and numerical alignment occur by approximately  $T = 153$  and  $T = 372$ , respectively, after which the Human and AI curves remain superposed. State  $B$  approaches  $p(a_0 | B) = 1$ , whereas the aligned policy in state  $A$  continues a slow displacement over almost the entire horizon and ends at  $p(a_0 | A) = 0.95958$ . It enters and remains within 0.01 of this terminal value only near  $T = 48638$ . Thus, numerical Human–AI alignment precedes common-policy stationarity by more than 48000 decision instants.





(a) Constant- $\eta$ , endogenous states ( $r_s = F$ ).



(b) Global diminishing- $\eta$ , exogenous states ( $r_s = T$ ),  $p_A = 0.30$ .

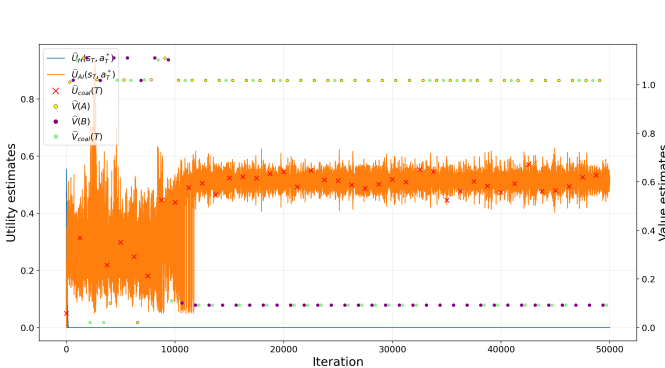
Figure 9: Representative predecision Human and AI probabilities assigned to  $a_0$  at  $H = 50000$ . The constant- $\eta$  configuration uses  $\eta = 0.025$  and  $(\alpha_{\text{agree}}, \alpha_{\text{disagree}}, \beta, \gamma) = (0.35, 0.51, 0.05, 0.10)$ . The global diminishing configuration uses  $\eta_T = 0.10/(1 + 0.002T)$  and  $(0.49, 0.85, 0.10, 0.50)$ .

The remaining selected policy trajectories confirm rather than alter this interpretation. Constant- $\eta$  exogenous learning reaches exact two-state alignment and becomes visually flat near the beginning of the horizon. The endogenous global diminishing winner reaches practical and numerical alignment by approximately  $T = 135$  and  $T = 201$ , and enters its terminal neighbourhood near  $T = 3238$ . Exact empirical updating aligns by  $T = 6$  under endogenous generation and by  $T = 11$  under exogenous generation. These plots are not reproduced because their informative transitions are compressed near the origin at this scale.

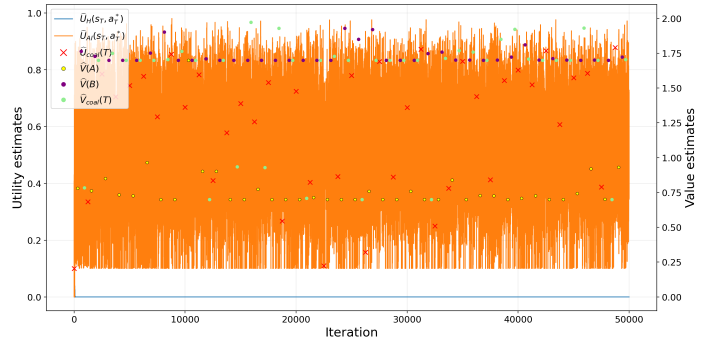
## 4.2 Utility-credit traces and value iterates

The constant- $\eta$  endogenous run assigns 99.926% of execution to the AI. Its final realized AI and coalition credit traces are 0.49787, while the finite-time value iterates end at  $\hat{V}(A) = 1.01852$  and  $\hat{V}(B) = 0.09259$ . The broad variation visible before the terminal segment reflects switches among realized state-action entries, not renewed Human-AI policy disagreement.

In the global diminishing exogenous run, AI execution reaches 99.988%. The final realized AI and coalition credit traces equal 0.44565, and the value iterates end at  $\hat{V}(A) = 0.78339$  and  $\hat{V}(B) = 1.85118$ . The differing value scale follows from the selected discount factor and realized transition path. As throughout this supplement, utility-credit traces and computational value iterates are diagnostics and do not replace the structural reward or alignment criterion.



(a) Constant- $\eta$ , endogenous states ( $r_s = F$ );  $\beta = 0.05$ ,  $\gamma = 0.10$ .



(b) Global diminishing- $\eta$ , exogenous states ( $r_s = T$ ),  $p_A = 0.30$ ;  $\beta = 0.10$ ,  $\gamma = 0.50$ .

Figure 10: Selected utility-credit traces and finite-time value iterates at  $H = 50000$ .

The nondisplayed utility-credit panels are retained in the complete repository. Exact empirical updating is essentially flat at this horizontal scale, while the remaining constant and global diminishing configurations reproduce the same path-dependent credit and value mechanisms already visible at shorter horizons.

## 4.3 Jensen-Shannon divergence and execution frequencies

In the constant- $\eta$  endogenous configuration,

$$(f_{\text{agree}}, f_{\text{ctx}}, f_{\text{disagree}}) = (0.99818, 0.00134, 0.00048),$$



and  $f_{AI} = 0.99926$ . The cumulative frequencies stabilize after a short initial transient, while the policy panel shows that the aligned common policies continue evolving for thousands of additional decisions.

The global diminishing exogenous trajectory yields

$$(f_{\text{agree}}, f_{\text{ctx}}, f_{\text{disagree}}) = (0.99946, 0.00050, 0.00004),$$

with  $f_{AI} = 0.99988$ . The almost immediate stabilization of the frequency curves coexists with a common state-A policy that remains mobile until near the terminal instant. This is the clearest long-horizon evidence that frequency-convergence equilibrium does not require immediate pointwise freeze-out of the aligned policy process.

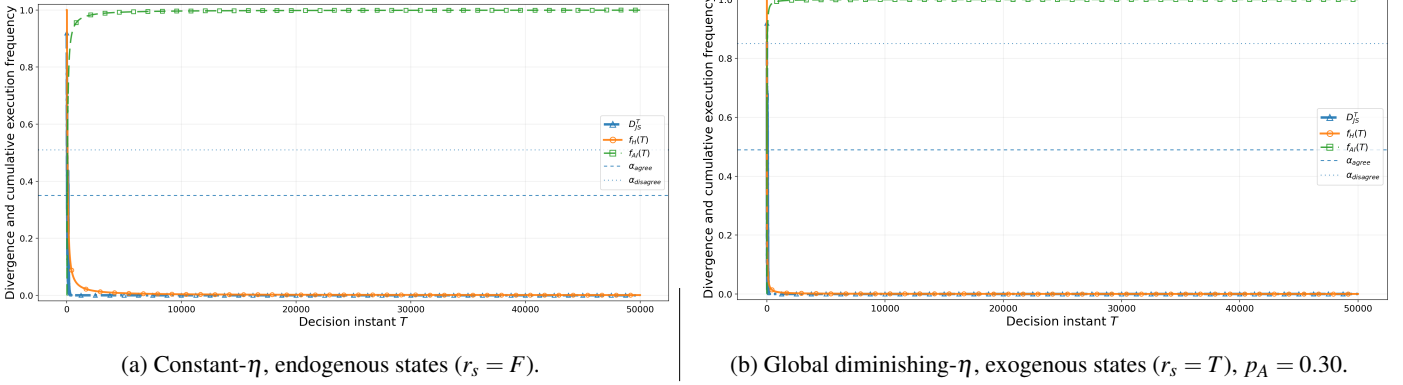


Figure 11: Jensen–Shannon divergence, delegation thresholds, and cumulative execution frequencies for the representative  $H = 50000$  configurations.

All six selected winners end with zero or numerically negligible state-conditioned divergence. Their agreement frequencies range from 0.99818 to 0.99996, and their AI execution frequencies range from 0.99926 to 0.99996. These selected configurations therefore represent AI-dominant frequency convergence, although the complete grid continues to distinguish the three update schedules.

#### 4.4 Horizon-level synthesis

Table 3: Two-state alignment over the complete  $H = 50000$  grid. Practical alignment requires  $D_{JS,\max}^{\text{final}} \leq 10^{-2}$ , and numerical alignment requires  $D_{JS,\max}^{\text{final}} \leq 10^{-6}$ .

| State generation | Learning schedule          | Practical | Numerical |
|------------------|----------------------------|-----------|-----------|
| Endogenous       | Constant- $\eta$           | 100.00%   | 100.00%   |
|                  | Global diminishing- $\eta$ | 96.67%    | 96.67%    |
|                  | Exact empirical            | 100.00%   | 100.00%   |
| Exogenous        | Constant- $\eta$           | 99.58%    | 97.50%    |
|                  | Global diminishing- $\eta$ | 70.28%    | 60.83%    |
|                  | Exact empirical            | 100.00%   | 100.00%   |

The complete grid confirms robust long-horizon alignment for constant gain and exact empirical updating. Constant- $\eta$  learning is practically aligned in every endogenous configuration and in 99.58% of the exogenous grid. Exact empirical updating achieves practical and numerical alignment in all evaluated configurations.

The principal residual dependence remains global diminishing learning under exogenous state generation. Practical alignment reaches 96.67% for  $p_A = 0.30$  and 95.00% for  $p_A = 0.70$ , but only 35.83% for  $p_A = 0.01$  and 27.50% for  $p_A = 0.99$ . Moreover, the overall exogenous practical proportion increases by only 0.69 percentage points between  $H = 30000$  and  $H = 50000$ . The pattern has therefore entered a long-horizon plateau rather than converging uniformly across the complete grid.

The mechanism is consistent with the distinction established in the previous horizons. Under global time decay, a rarely visited state may receive informative updates only after the common step size has become small. Constant gain preserves responsiveness to late observations, whereas exact empirical updating scales each agent–state component by its own effective local count. Consequently,  $H = 50000$  validates persistent alignment and stabilized execution frequencies, but retains a measurable schedule- and coverage-dependent limitation. It should be interpreted as long-horizon consolidation, not as universal saturation.

## 5 Computational validation of entropy-based Human–AI alignment

Taken together, Chapters I and II provide a systematic finite-horizon validation of Scenario 1, Human arbitration. The experiments cover the exploratory transient, early learning, emerging stabilization, and long-horizon consolidation through the independently executed horizons

$$H \in \{200, 1000, 5000, 15000, 30000, 50000\}.$$

The results support the central computational hypothesis that the Jensen–Shannon divergence provides an operational signal of Human–AI policy alignment. Across the representative trajectories, the contraction of the state-conditioned divergence accompanies the reduction of the differences between the Human and AI policy distributions. When the policies become equal in both states,  $D_{JS}$  approaches zero; when one state remains insufficiently learned, the corresponding state-conditioned divergence continues to identify the residual discrepancy. The divergence therefore distinguishes genuine two-state alignment from apparent convergence caused by the terminal observation of only one well-sampled state.

This empirical result reinforces the information-theoretic role assigned to  $D_{JS}$  in the main paper [?]. Its symmetry, boundedness, and normalization make it suitable for comparing Human and AI policies on a common scale and for determining whether a decision lies in the agreement, contextual, or disagreement region. The simulations show that this classification remains interpretable across learning-rate schedules, state-generation mechanisms, and horizons.

The evidence must nevertheless be interpreted with the theoretical distinction established in Part I. The Jensen–Shannon divergence is an alignment statistic and a delegation signal; it is not a welfare functional, an optimality criterion, or a Lyapunov function for the policy recursion. Alignment is generated by repeated execution, asymmetric incorporation of executed actions, and the corresponding policy-adaptation dynamics. The role of  $D_{JS}$  is to measure the resulting inter-policy discrepancy and to connect that discrepancy to the delegation architecture.

The complete horizon sequence also supports the frequency-convergence equilibrium formulated in Part I. In the selected long-horizon configurations, the agreement region occupies almost the entire realized path and execution is overwhelmingly AI-owned. Under Human arbitration, this constitutes an AI-dominant equilibrium in empirical occupation frequency: the Human retains structural override authority in disagreement, while repeated learning progressively places most decisions inside the agreement region. Stability is therefore behavioural and dynamic rather than a one-shot Nash, Stackelberg, or best-response fixed point.

The computational results further clarify that policy alignment, frequency stabilization, and common-policy stationarity are distinct properties. Human and AI policies may become numerically identical while their common state-conditioned distribution continues to evolve. Likewise, agreement and execution frequencies may stabilize long before the aligned common policy becomes visually flat. This confirms that the equilibrium concept concerns stabilization of empirical regime frequencies and does not require immediate pointwise freeze-out of every policy trajectory.

The learning-rate schedules determine how robustly this mechanism operates across the evaluated grid. Exact empirical updating produces the strongest two-state alignment because its step size depends on the local effective count of each agent–state component. Constant gain retains responsiveness to observations arriving late in the horizon. Global diminishing gain remains more sensitive to extreme exogenous state imbalance because a rarely visited state may begin receiving informative observations only after the global step size has become small. These residual cases delimit the scope of the computational validation rather than contradicting the alignment hypothesis.

Accordingly, the simulations do not establish universal pointwise convergence for every admissible parameter configuration. They provide the more precise result that, within the evaluated finite-state and finite-action setting,  $D_{JS}$  reliably tracks Human–AI policy alignment, the two-threshold architecture translates that signal into coherent delegation regimes, and repeated executed-learning produces stable agreement and AI-dominant execution frequencies over broad regions of the parameter grid.

The two supplementary chapters therefore reinforce the theoretical foundation of Neo-Game Theory established in Part I. Hybrid stability can emerge without classical best-response optimization: Human and AI retain separate policies and structural utilities, execution authority is selected endogenously under Virtual Nature, and equilibrium is expressed through the long-run empirical behaviour of the delegation–learning process.

## References

- [1] S. A. Sepúlveda-Fontaine and J. M. Amigó, “An Entropy-Based Framework for Hybrid Coalitions in Game Theory—Part I: Human Arbitration,” *Entropy*, vol. 28, no. 4, Art. 473, 2026. <https://doi.org/10.3390/e28040473>.